THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL UNIT TEST SET 1

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home: Earth, a Unique Life-sustaining Planet** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Assertion (A): The image formed by a plane mirror is virtual. Reason (R): It cannot be obtained on a screen. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Assertion (A): The shadow of a vertical stick is shortest at noon. Reason (R): At noon the Sun is highest in the sky. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Assertion (A): Removing top carnivores from a forest can harm the whole ecosystem. Reason (R): Herbivore numbers increase and overgraze the vegetation. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Assertion (A): The Earth's inner core is solid even though it is extremely hot. Reason (R): The very high pressure at the centre prevents it from melting. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. The angle between a ray and a plane mirror is 35 degrees. If the mirror is rotated by 15 degrees so that the angle of incidence increases, the new angle between the incident and the reflected ray is: 1

(a) 70 degrees **(b)** 110 degrees **(c)** 100 degrees **(d)** 140 degrees

6. A pole of height 3 m has a shadow of 4 m and a nearby tree has a shadow of 20 m at the same time. The height of the tree is: 1

(a) 20 m **(b)** 26 m **(c)** 10 m **(d)** 15 m

7. A food chain has producers with 50000 J. If it has 5 levels (producers, primary, secondary, tertiary, quaternary), the energy at the quaternary level is: 1

(a) 500 J **(b)** 5 J **(c)** 50 J **(d)** 0.5 J

8. The core of the Earth has a radius about 3400 km and the Earth about 6400 km. The ratio of the core radius to the Earth radius is about: 1

(a) 1.88 **(b)** 0.34 **(c)** 0.53 **(d)** 0.64

9. Light incident on a smooth metal surface at 90 degrees to the surface is reflected at: 1

(a) 180 degrees
(b) 0 degrees
(c) 45 degrees
(d) 90 degrees to the surface (back along normal)

10. The Moon appears the same size as the Sun in the sky although the Sun is about 400 times larger because the Sun is also: 1

(a) dimmer **(b)** about 400 times farther
(c) smaller **(d)** closer

11. Which of the following is a good example of mutualism? 1

(a) bees and flowers **(b)** mosquito and human
(c) tapeworm and human **(d)** tiger and deer

12. A glacier's melting adds water to the oceans. If all land ice melted, the effect on coastal cities would be: 1

(a) no change **(b)** flooding by rising sea levels
(c) cooler climate **(d)** drought on the coast

13. Assertion (A): A concave mirror is used as a shaving mirror. Reason (R): It forms an enlarged erect image of the face when held close. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. Assertion (A): A leap year has 366 days. Reason (R): The Earth takes about 365.25 days to complete one revolution around the Sun. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. Assertion (A): Energy flow in an ecosystem is unidirectional. Reason (R): Energy lost as heat cannot be reused by producers. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A ray incident on a mirror at 30 degrees to the surface is reflected. If the mirror turns through 10 degrees, find the possible new angle of incidence. 2

17. Why does a calendar need a leap-day rule with exceptions for century years? Give an example with 1900 and 2000. 2

18. Calculate the energy available to the primary consumer, secondary and tertiary if producers have 100000 J (10 per cent rule). 2

19. The core of the Earth accounts for 15 per cent of the Earth's volume. If the total volume is about 1.08 x 10^12 cubic km find the volume of the core (approx). 2

20. Why does a plane mirror show an image of the same size even though the image looks 'far away'? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Two plane mirrors are inclined at an angle theta. If 7 images are formed, find theta. 3

22. At a place the Sun is directly overhead at noon on 21 March. A pole of height 5 m is placed there. Describe the shadow at noon and at 3 pm. 3

23. A pond has 1000 kg of algae. By the 10 per cent biomass rule, find the maximum mass of tertiary consumers it can support. 3

24. Describe how Earth's layers formed by differentiation and what evidence supports a liquid outer core. 3

OR

A student uses a concave mirror as a make-up mirror and sees an enlarged face. When she moves away, the face appears inverted. Explain. 3

25. Why do countries have different time zones? How many hours of difference per 15 degrees of longitude? 3

OR

Describe the effects of eutrophication in a lake in three stages. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe the formation of the image of an object placed in front of a concave mirror at six positions. For the object at 30 cm from a mirror of focal length 10 cm state the position, nature and size. 5

OR

Describe a long-term investigation to record the Sun's noon shadow length and direction through the year and explain the expected results with the solstices and equinoxes. 5

27. The population of lions in a forest dies due to disease. Trace the consequences through the food web with at least five steps in the ecosystem and the final outcome. 5

OR

The Earth's radius is 6400 km. Using the crust 35 km, mantle 2900 km, outer core 2200 km, find the radius of the inner core, the percentage of radius for each layer and comment on which layer is thickest. 5

END OF QUESTION PAPER — Total: 50 Marks